

A HERS for Carbon?

While “carbon emissions,” “greenhouse gases,” and “carbon footprint” have become well-known terms of widespread concern, the notion of carbon-neutral building might have more people turning to the Internet for a definition.

A conversation about HERS and zero-carbon homes between Kelly Parker, CEO of Guaranteed Watts Savers (GWS) in Oklahoma and Texas, and current president of the Residential Energy Services Network (RESNET), and Jane Talkington, a proponent of passive design strategies and the sustainability scholar of Trifecta-Consulting.com:

Jane: RESNET is in discussions about carbon-neutral building metrics, so let's begin by putting some buzzwords in perspective. While “carbon emis-

sions,” “greenhouse gases,” and “carbon footprint” have become well-known terms of widespread concern, the notion of carbon-neutral building might

have more people turning to the Internet for a definition. It would seem that a lot of people are talking about the problem, but there is less conversation about the solution. We know buildings are responsible for almost half (48%) of U.S. annual energy consumption and greenhouse gas emissions. Where do you think your industry is now, and in what direction is it headed?

Kelly: There has definitely been a change in momentum this year. There

is less controversy about the existence of global warming and much more talk about ways to mitigate it. That talk is beginning to generate solutions. We've

already seen the design sector launch initiatives like www.architecture2030.org. This initiative is intended to increase the energy efficiency of buildings.

But the initiative

founder, the American Institute of Architects (AIA), exerts an influence mostly on the design of commercial buildings. I believe RESNET has a role to play in addressing the home-building industry. We are in talks with the Architecture2030 group about developing energy efficiency guidelines for homes.

Jane: I've read that the RESNET board of directors has shown some leadership in carbon emission reductions. At their annual conference, RESNET offset their carbon footprint by buying carbon offsets. RESNET is also in the early stages of developing a new scale that complements the Home Energy Rating System (HERS) index—a scale that would measure the carbon emissions from a home. What does the RESNET board hope to accomplish with this new scale?

Kelly: As the adage goes, we can't control what we can't measure. The

current HERS index is only for heating, cooling, water heating, lights, and appliances but not plug loads. HERS does not consider everything. For example, HERS doesn't consider the source of energy used in a home and how efficiently or inefficiently it is created; and it doesn't consider water use. The solution is to design a new index that measures everything.

Jane: I'm not confident that we can control something even after we measure it.

Kelly: True, but we need to establish a reference point so we can gauge future progress.

Jane: Are you talking about calculating carbon emissions on new homes or on existing homes?

Kelly: For starters, new homes must be designed and built to use as little energy and other resources as possible. HERS raters are also involved with energy audits and retrofitting existing buildings, so we have an opportunity to lower overall emissions if we build wisely while we correct past mistakes. We know we have to reduce carbon emissions, and we can do this immediately by managing demand for electricity. Ultimately, a low-carbon economy would result in low or acceptable carbon emissions, but that will only happen when we get all or most of our energy from renewables.

Jane: Carbon-neutral building is controversial for many reasons. Carbon offsetting is fraught with complications. These range from determining how long it should take to completely mitigate present-day emissions to evaluating the ecological imbalance caused in the soil and water systems by planting huge swathes of forests. Many firms use the money from selling carbon offsets to invest in renewable technology research. Some carbon offset firms are intent on maintaining their own voluntary standards to avoid regulation.

Opinions are divided as to the benefit of the offset industry. Many



JANE TALKINGTON

experts believe that offsetting allows our world society to avoid the real issue of reducing its carbon footprint by just buying off bad behavior instead of making fundamental lifestyle changes. The solution may be somewhere in the middle: reduce your carbon output and mitigate what can't be reduced through lifestyle choices.

Kelly: Those are valid points. I wish there was an easy answer, but there is not. Our task is to look at the impact that operating a building has on the environment as a whole, and figure out feasible ways to offset that impact. The real opportunity is in the planning stages; this is where we can design out and engineer out high loads. We have a RESNET network in place that can initiate substantial changes by incremental progress, one home at a time.

Jane: What do you propose to measure with your new index?

Kelly: Right now our rating is a site-dependent modified end-use load. The source energy and carbon footprint should be tied directly to the regional power plant. This is what carbon offsets are all about—offsetting the emissions of the local power plant.

Jane: You are talking just about the carbon impact of operating a home. What about the emissions created in the initial construction of the home? A hand-built adobe home and an insulated concrete form (ICF) home would have drastically different carbon footprints in the construction phase. Wouldn't we have to consider how long the structure remains in use to determine if a high carbon footprint justifies its existence by increasing energy efficiency and longevity?

Kelly: Because we don't know exactly how long a structure will stand, we can't measure whether the initial carbon emissions released are cost-effective. We can only measure the home's annual operational load. In England you can pick up a bag of potato chips, and the carbon footprint of the chips is printed on the bag. Surveys reveal that 50% of

consumers use the carbon footprint in making purchasing decisions. We'll have to wait until building materials in America are labeled with a carbon footprint before we can develop an index that deals with the construction aspect. For now, RESNET can only look at the operational cost of a home.

Jane: The whole concept of buying carbon offsets seems riddled with questionable theories, wouldn't you say? Every time we build, we have an impact on resources and on the global climate. To pretend that this goes away when we plant a forest somewhere seems sim-

There are two elements to carbon offsetting. One is planting trees to sequester or soak up the carbon emissions released by a certain activity. The other side, the side that makes the most sense to me, is local power generation by the use of renewables.

plistic and naïve. I think carbon offsets are a form of environmental bulimia, where we pollute, then unpollute, and consider it even.

Kelly: There are two elements to carbon offsetting. One is planting trees to sequester or soak up the carbon emissions released by a certain activity. This assumes that there are limitless supplies of land and water to support a forest for the next 50 years in order to compensate for the carbon emissions produced by an airplane flight today. The other side, the side that makes the most sense to me, is local power generation by the use of renewables. This could mean substituting wind energy for fossil fuel energy, or using photovoltaics on-site.

Jane: Don't forget to promote passive strategies, like passive-solar heating from southern windows and solar shading to prevent unwanted heat gain in the summer. These are so easy; they require only a little money but a lot of thought.

Kelly: Exactly, but strategies like these require design changes, and raters rarely deal with changes in the design of a building. In the future, raters will have opportunities to be the catalysts for these design changes. There are really amazing opportunities to be realized if we just roll up our sleeves and get to work. New software is available to help us do this. EPA has developed Target Finder to determine energy performance, and Portfolio Manager to measure the CO₂ release of commercial buildings. The Green Building Studio's Carbon Neutral Design Tool is in beta version.

Jane: This gets us back to the Zero Energy Home concept. Isn't that enough of a solution? Why go beyond zero energy building into the complex world of carbon emissions?

Kelly: I don't think we have the luxury of choice in the matter. I think we have to address carbon emissions in the broader sense, because of the catastrophic future an unchecked carbon economy might bring.

Jane: I was visiting with Mack Caldwell the other day about Ideal Home's Zero Energy Home prototype. Mack was the architect of that home. He shared a ton of lessons learned but I think the point that fascinated me the most was the influence of user behavior. That house was specifically designed to operate at 76°F in the summer but the occupant set the thermostat on 70°F. This resulted in the actual energy use being about twice the amount from April to September than was predicted by the modeling software.

Kelly: Did the home builder provide adequate buyer education?

Jane: Yes, but evidently consumers nod in agreement but then do what they want anyway. Research shows us over and over that user behavior can negate so much of the technological advances. We have to remember to influence the consumers.

Kelly: We've got to do a better job explaining how a high-performance house works to the end customer. I hear a lot of new catch phrases in the average person's conversation that just weren't there a year ago, and these catch phrases need explaining.

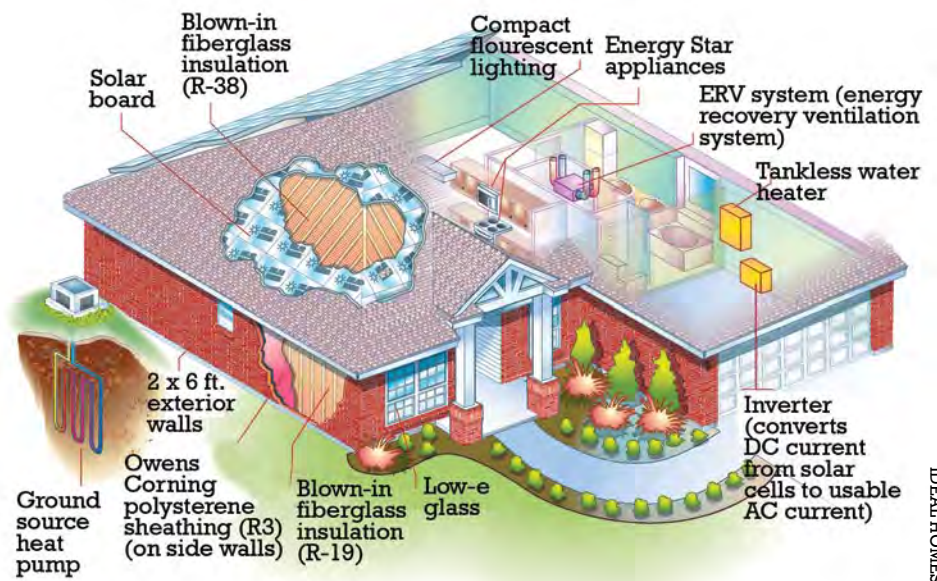
Jane: Me too. Mack told me that Ideal Homes is currently working on developing a zero-carbon emissions house. I'm always proud of them for learning and pushing the boundaries in a conservative way. They do the entire industry a huge favor when they share what they have learned.

Jane: What advice would you give to HERS raters who are on the front lines of this carbon-neutral issue?

Kelly: I would advise them to work on developing a holistic view of all of our nation's energy and resource use, and to continue to promote strategies to use energy and resources more wisely. I'll be honest—the industry is evolving so rapidly that raters will need to keep themselves educated on emerging issues. Raters have technical minds and are good at translating complex topics into meaningful ideas for those less technically savvy. They are also a trusted source of advice, which makes them an excellent source to introduce change—even design changes.

Jane: Let's say we build a zero energy house with a low carbon emission rating; but if the homeowner commutes in a Hummer 300 miles each day, what progress is that? The right building in the wrong place is the wrong building.

Kelly: True, but that is a developer's issue and falls in the arena of sustain-



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able urban planning. It is also a cultural issue. As carbon-neutral building and carbon footprinting become mainstream concepts, we'll see more lifestyle changes conducive to a low-carbon economy.

Jane: Alex Steffan, editor of *World-Changing*, writes that suburban communities emit 11 tons of CO₂ per home, whereas residential units in dense areas emit 9 tons or less per home. This leads me to conclude that even if we can measure carbon emissions from a home's operation, overall emission rates are greatly affected by the housing density of the city in which the house is built, the availability of mass transportation, and even the lifestyle of the occupant. It is an extremely complex issue to address, isn't it?

Kelly: Yes, in reality, solving one question leads to twenty more! Energy is a moving target. What I see happening is a convergence of many disciplines to find that optimal solution. I didn't say perfect, merely optimal, given what we know today and can do.

Jane: Agreed. The only place we can start is where we are now; and where is that? What percentage of new homes built nationwide are Energy Star certified?

Kelly: There are 15 states that each have a penetration level of 12% or better. In 2006, nearly 200,000 new homes nationwide earned Energy Star certification, bringing the total of Energy Star-certified homes in the United States to 750,000. This is a great start but we need to see Energy Star become a standard fixture.

Jane: That leaves a lot of room for improvement, especially considering how few of those homes were also green certified and/or zero energy homes. It seems that the home-building industry is not exactly blazing the trail toward sustainability. We have the exceptions, like Ideal Homes in my home state of Oklahoma; they are the production builders who built the first zero energy home in the nation for under \$200,000.

Kelly: What we do is identify the champions who are shifting the industry. One home may not have much of an impact, but it sets an example. The first domino has got to fall for the others to go. Advances in the home-building industry take place in predictable steps. First, builders tackle energy efficiency and verify their success with third-party certification from Energy Star providers. In some markets, such as Las Vegas, we are seeing close to a 50% penetration rate of Energy Star homes. Next, builders incorporate green building by following

a rating system such as LEED certification. The more progressive builders accept the challenge of building a zero energy home. The ultimate task before the industry now is to create a new construction standard so that all buildings will eventually be carbon neutral. Actually, this is the ultimate task before every society on the planet.

Jane: All these advances are voluntary; nothing is mandated by code; and there is no consumer market driving innovation, so the evolution is slow. Until Energy Star, carbon emissions, or green building, becomes law, cultural preferences will prevail. Unfortunately, I can't say that our culture is very building savvy or energy conscious.

Kelly: But Jane! That is why we're here! Every movement begins with catalysts. The leadership of RESNET has the foresight to look for solutions sooner rather than later. The raters have the skills to take technical solutions to the market, so it's really just a matter of how quickly the market can

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accept the changes and new technologies. The most challenging part is getting the market to accept new philosophies and new definitions of what high-performance housing is.

Jane: I think we've stirred the pot enough to create some dialogue. Which Web sites would you suggest raters visit to find in-depth coverage of carbon-neutral building issues?

Kelly: An easy place to start is Wikipedia.com, because they frame the basic terminology so well. The BedZed development in the U.K. offers a Toolkit for Carbon Neutral Developments. A 13-page extract of this toolkit can be viewed on-line at www.bioregional.com/programme_projects/ecohous_prog/bedzed/bz_carbon.htm and purchased by

e-mailing jennieorgan@bioregional.com. The U.S. Green Building Council (USGBC) has information on emerging issues surrounding carbon-neutral building. Go to www.usgbc.org and type "carbon neutral" in the search box at the top of the page. You can also go to www.greenbuildingstudio.com to watch a video about software that can calculate the carbon neutrality of a building. This is a great time to jump in the conversation, because we only find answers after we ask the questions. RESNET maintains a RESblog where raters can share their thoughts about carbon footprinting, go to www.natresnet.org/resblog. And of course you can write on the *Home Energy* blog at www.homeenergy.org/blog.php.

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